

Functional Programming

Data Wrangling in R

Functional Programming

“R, at its heart, is a functional programming (FP) language. This means that it provides many tools for the creation and manipulation of functions.” - [Hadley Wickham](#)

Don't need to write for-loops! - check this [video](#).

Iterative work

Allows you to flexibly iterate functions to multiple elements of a data object!

Useful when you want to apply a function to:

- * lots of columns in a tibble
- * multiple tibbles
- * multiple data files

base R **apply** functions

Works really simply for all columns.

```
sapply(mtcars, FUN = round)
```

```
##      mpg  cyl  disp  hp  drat  wt  qsec  vs  am  gear  carb
## [1,]  21    6  160 110    4    3   16   0   1    4    4
## [2,]  21    6  160 110    4    3   17   0   1    4    4
## [3,]  23    4  108  93    4    2   19   1   1    4    1
## [4,]  21    6  258 110    3    3   19   1   0    3    1
## [5,]  19    8  360 175    3    3   17   0   0    3    2
## [6,]  18    6  225 105    3    3   20   1   0    3    1
## [7,]  14    8  360 245    3    4   16   0   0    3    4
## [8,]  24    4  147  62    4    3   20   1   0    4    2
## [9,]  23    4  141  95    4    3   23   1   0    4    2
## [10,] 19    6  168 123    4    3   18   1   0    4    4
## [11,] 18    6  168 123    4    3   19   1   0    4    4
## [12,] 16    8  276 180    3    4   17   0   0    3    3
## [13,] 17    8  276 180    3    4   18   0   0    3    3
## [14,] 15    8  276 180    3    4   18   0   0    3    3
## [15,] 10    8  472 205    3    5   18   0   0    3    4
## [16,] 10    8  460 215    3    5   18   0   0    3    4
## [17,] 15    8  440 230    3    5   17   0   0    3    4
## [18,] 32    4    79  66    4    2   19   1   1    4    1
## [19,] 30    4    76  52    5    2   19   1   1    4    2
## [20,] 34    4    71  65    4    2   20   1   1    4    1
## [21,] 22    4   120  97    4    2   20   1   0    3    1
## [22,] 16    8   318 150    3    4   17   0   0    3    2
## [23,] 15    8   304 150    3    3   17   0   0    3    2
```

Base R **apply** functions specific columns

Harder to use for specific columns.

```
rounded_mtcars <- mtcars %>% select(dispatch, drat, wt, qsec) %>%  
  apply(FUN = round)  
bind_cols(rounded_mtcars, select(mtcars, -c(dispatch, drat, wt, qsec)))
```

	dispatch	drat	wt	qsec	mpg	cyl	hp	vs	am
## Mazda RX4	160	4	3	16	21.0	6	110	0	1
## Mazda RX4 Wag	160	4	3	17	21.0	6	110	0	1
## Datsun 710	108	4	2	19	22.8	4	93	1	1
## Hornet 4 Drive	258	3	3	19	21.4	6	110	1	0
## Hornet Sportabout	360	3	3	17	18.7	8	175	0	0
## Valiant	225	3	3	20	18.1	6	105	1	0
## Duster 360	360	3	4	16	14.3	8	245	0	0
## Merc 240D	147	4	3	20	24.4	4	62	1	0
## Merc 230	141	4	3	23	22.8	4	95	1	0
## Merc 280	168	4	3	18	19.2	6	123	1	0
## Merc 280C	168	4	3	19	17.8	6	123	1	0
## Merc 450SE	276	3	4	17	16.4	8	180	0	0
## Merc 450SL	276	3	4	18	17.3	8	180	0	0
## Merc 450SLC	276	3	4	18	15.2	8	180	0	0
## Cadillac Fleetwood	472	3	5	18	10.4	8	205	0	0
## Lincoln Continental	460	3	5	18	10.4	8	215	0	0
## Chrysler Imperial	440	3	5	17	14.7	8	230	0	0
## Fiat 128	79	4	2	19	32.4	4	66	1	1
## Honda Civic	76	5	2	19	30.4	4	52	1	1
## Toyota Corolla	71	4	2	20	33.9	4	65	1	1
## Toyota Corona	120	4	2	20	21.5	4	97	1	0

Working **across** multiple columns

`across` allows us to perform functions on specific columns more easily.

```
mutate(across(which_columns, which function or operation))
```

```
head(mtcars, 2)
```

```
##           mpg  cyl  disp  hp  drat    wt    qsec  vs  am  gear
## Mazda RX4    21   6  160 110   3.9  2.620 16.46  0  1     4
## Mazda RX4 Wag 21   6  160 110   3.9  2.875 17.02  0  1     4
##           carb
## Mazda RX4      4
## Mazda RX4 Wag  4
```

```
mtcars %>%
  mutate(across(.cols = c(disp, drat, wt, qsec), round)) %>%
  head(2)
```

```
##           mpg  cyl  disp  hp  drat  wt  qsec  vs  am  gear  carb
## Mazda RX4    21   6  160 110    4  3    16  0  1    4    4
## Mazda RX4 Wag 21   6  160 110    4  3    17  0  1    4    4
```

Using **across** with arguments

If you want to use arguments, then you need to use the `~` and `.x` (or `.`) as a placeholder for what you will be passing into the function.

See [purrr-shortcuts](#) and [this link about . and .x](#).

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(.x, digits = 1))) %>%  
  head(n = 2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear  
## Mazda RX4      21   6  160 110   3.9 2.6 16.5  0  1    4  
## Mazda RX4 Wag  21   6  160 110   3.9 2.9 17.0  0  1    4  
##  
##           carb  
## Mazda RX4      4  
## Mazda RX4 Wag  4
```

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(., digits = 1))) %>%  
  head(n = 2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear  
## Mazda RX4      21   6  160 110   3.9 2.6 16.5  0  1    4  
## Mazda RX4 Wag  21   6  160 110   3.9 2.9 17.0  0  1    4  
##  
##           carb  
## Mazda RX4      4  
## Mazda RX4 Wag  4
```

Writing functions

Why write your own functions?

- Cut down on repetitive code (easier to fix things!)
- Organize code into manageable chunks
- Avoid running code unintentionally

Functions in R

To create a function we can use the `function` function and specify what input the function will take and what it will do to it.

```
my_function <- function(x){x + 1}  
my_function
```

```
## function (x)  
## {  
##     x + 1  
## }
```

```
my_data <- c(2,3,4)  
my_function(x = my_data)
```

```
## [1] 3 4 5
```

```
my_function(my_data)
```

```
## [1] 3 4 5
```

purrr is also a super helpful package!

“Designed to make your functions purrr.”

`dplyr` is designed for data frames `purrr` is designed for vectors

The `purrr` package can be very helpful!

- <https://purrr.tidyverse.org/>
- <https://github.com/rstudio/cheatsheets/raw/master/purrr.pdf>
- <https://jennybc.github.io/purrr-tutorial/>

purrr main functions

map and map_* (and modify)

- **applies function** to each element of an vector or object (map returns a list, modify returns the same object type) - map_df will output a dataframe

Fancier Options

map2 and map2_*

- applies function to each element of **two** vectors or objects
- an example would be improving multiple files with a vector of different delimiters

pmap and pmap_* - applies function to each element of **3+** vector or objects (requires a list for input)

the *_* options specify the type of data output

map_df and across

Both `map_df` and `across` can do the same thing, `map_df` is simpler, but `across` keeps rownames.

```
mtcars %>%  
  map_df(round, digits = 1) %>%  
  head(n = 2)
```

```
## # A tibble: 2 × 11  
##   mpg   cyl  disp    hp  drat    wt   qsec    vs  am  
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1    21     6   160   110   3.9   2.6  16.5     0     1  
## 2    21     6   160   110   3.9   2.9  17.0     0     1  
## # i 2 more variables: gear <dbl>, carb <dbl>
```

```
mtcars %>%  
  mutate(across(.cols = everything(), ~ round(.x, digits = 1))) %>%  
  head(n = 2)
```

```
##           mpg cyl disp  hp drat   wt  qsec vs am gear  
## Mazda RX4    21   6  160 110   3.9  2.6  16.5  0  1    4  
## Mazda RX4 Wag 21   6  160 110   3.9  2.9  17.0  0  1    4  
##           carb  
## Mazda RX4      4  
## Mazda RX4 Wag  4
```

purrr apply function to some columns like across

Using `modify_if()` (or `map_if()`), we can specify what columns to modify

```
head(as_tibble(iris), 3)
```

```
## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1         5.1         3.5           1.4           0.2 setosa
## 2         4.9         3             1.4           0.2 setosa
## 3         4.7         3.2           1.3           0.2 setosa
```

```
as_tibble(iris) %>%
  modify_if(is.numeric, round) %>%
  head(3)
```

```
## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1         5         4             1           0 setosa
## 2         5         3             1           0 setosa
## 3         5         3             1           0 setosa
```

Lists

What is a list?

- Lists are the most flexible/"generic" data class in R
- Can be created using list()
- Can hold vectors, strings, matrices, models, list of other lists, lists upon lists!

```
mylist <- list(  
  letters = c("A", "b", "c"),  
  numbers = 1:3,  
  matrix(1:25, ncol = 5),  
  iris  
)
```


List Structure

```
head(mylist)
```

```
## $letters
## [1] "A" "b" "c"
##
## $numbers
## [1] 1 2 3
##
## [[3]]
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    6   11   16   21
## [2,]    2    7   12   17   22
## [3,]    3    8   13   18   23
## [4,]    4    9   14   19   24
## [5,]    5   10   15   20   25
##
## [[4]]
##      Sepal.Length Sepal.Width Petal.Length Petal.Width
## 1             5.1          3.5          1.4          0.2
## 2             4.9          3.0          1.4          0.2
## 3             4.7          3.2          1.3          0.2
## 4             4.6          3.1          1.5          0.2
## 5             5.0          3.6          1.4          0.2
## 6             5.4          3.9          1.7          0.4
## 7             4.6          3.4          1.4          0.3
## 8             5.0          3.4          1.5          0.2
## 9             4.4          2.9          1.4          0.2
## 10            4.9          3.1          1.5          0.1
## 11            5.4          3.7          1.5          0.2
```

List referencing

```
mylist[1] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```

```
mylist["letters"] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```

List referencing

```
mylist[[1]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

```
mylist[["letters"]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

... Why lists?

List: `split()` a dataset

We can create a list by splitting up a dataframe. We will use `mtcars`.

```
head(mtcars)
```

```
##           mpg  cyl  disp  hp  drat    wt    qsec  vs  am
## Mazda RX4      21.0    6   160  110  3.90  2.620  16.46  0  1
## Mazda RX4 Wag  21.0    6   160  110  3.90  2.875  17.02  0  1
## Datsun 710      22.8    4   108   93  3.85  2.320  18.61  1  1
## Hornet 4 Drive  21.4    6   258  110  3.08  3.215  19.44  1  0
## Hornet Sportabout 18.7    8   360  175  3.15  3.440  17.02  0  0
## Valiant        18.1    6   225  105  2.76  3.460  20.22  1  0
##           gear carb
## Mazda RX4      4    4
## Mazda RX4 Wag  4    4
## Datsun 710      4    1
## Hornet 4 Drive  3    1
## Hornet Sportabout 3    2
## Valiant        3    1
```

List: `split()` the dataset by `cyl`

```
mtcars_split <-mtcars %>% split(.$cyl)
str(mtcars_split)
```

```
## List of 3
## $ 4:'data.frame':  11 obs. of  11 variables:
##   ..$ mpg : num [1:11] 22.8 24.4 22.8 32.4 30.4 33.9 21.5 27.3 26 30.4 ...
##   ..$ cyl : num [1:11] 4 4 4 4 4 4 4 4 4 4 4 ...
##   ..$ disp: num [1:11] 108 146.7 140.8 78.7 75.7 ...
##   ..$ hp  : num [1:11] 93 62 95 66 52 65 97 66 91 113 ...
##   ..$ drat: num [1:11] 3.85 3.69 3.92 4.08 4.93 4.22 3.7 4.08 4.43 3.77 ...
##   ..$ wt  : num [1:11] 2.32 3.19 3.15 2.2 1.61 ...
##   ..$ qsec: num [1:11] 18.6 20 22.9 19.5 18.5 ...
##   ..$ vs  : num [1:11] 1 1 1 1 1 1 1 1 0 1 ...
##   ..$ am  : num [1:11] 1 0 0 1 1 1 0 1 1 1 ...
##   ..$ gear: num [1:11] 4 4 4 4 4 4 3 4 5 5 ...
##   ..$ carb: num [1:11] 1 2 2 1 2 1 1 1 2 2 ...
## $ 6:'data.frame':  7 obs. of  11 variables:
##   ..$ mpg : num [1:7] 21 21 21.4 18.1 19.2 17.8 19.7
##   ..$ cyl : num [1:7] 6 6 6 6 6 6 6
##   ..$ disp: num [1:7] 160 160 258 225 168 ...
##   ..$ hp  : num [1:7] 110 110 110 105 123 123 175
##   ..$ drat: num [1:7] 3.9 3.9 3.08 2.76 3.92 3.92 3.62
##   ..$ wt  : num [1:7] 2.62 2.88 3.21 3.46 3.44 ...
##   ..$ qsec: num [1:7] 16.5 17 19.4 20.2 18.3 ...
##   ..$ vs  : num [1:7] 0 0 1 1 1 1 0
##   ..$ am  : num [1:7] 1 1 0 0 0 0 1
##   ..$ gear: num [1:7] 4 4 3 3 4 4 5
##   ..$ carb: num [1:7] 4 4 1 1 4 4 6
## $ 8:'data.frame':  14 obs. of  11 variables:
```

List: model on each

```
mtcars %>%  
  split(.$cyl) %>% # creates split of data for each unique cyl value  
  map(~lm(mpg ~ wt, data = .)) %>% # apply linear model to each  
  map(summary) %>%  
  map_dbl("r.squared")
```

```
##           4           6           8  
## 0.5086326 0.4645102 0.4229655
```

Use lists to iterate through multiple files!

This comes up a lot in data cleaning and also when reading in multiple files!

```
library(here)
library(readr)
file_list <- list.files(here::here("data/iris/"), pattern = "*.csv")

file_list <- paste0(here::here("data/iris/"), file_list)
file_list
```

```
## [1] "/Users/carriewright/Documents/GitHub/Teaching/SISBID_Data-Wrangling/data/iris/iris_q1.csv"
## [2] "/Users/carriewright/Documents/GitHub/Teaching/SISBID_Data-Wrangling/data/iris/iris_q4.csv"
## [3] "/Users/carriewright/Documents/GitHub/Teaching/SISBID_Data-Wrangling/data/iris/iris_q5.csv"
```

```
multifile_data <- file_list %>%
  map(read_csv)
```


Reading in multiple files -First item

```
multifile_data[[1]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Reading in multiple files - 2nd item

```
multifile_data[[2]]
```

```
## # A tibble: 150 × 1
##   Sepal.Length:Sepal.Width:Petal.Length:Petal.Width:Speci...1
##   <chr>
## 1 5.1:3.5:1.4:0.2:setosa
## 2 4.9:3:1.4:0.2:setosa
## 3 4.7:3.2:1.3:0.2:setosa
## 4 4.6:3.1:1.5:0.2:setosa
## 5 5:3.6:1.4:0.2:setosa
## 6 5.4:3.9:1.7:0.4:setosa
## 7 4.6:3.4:1.4:0.3:setosa
## 8 5:3.4:1.5:0.2:setosa
## 9 4.4:2.9:1.4:0.2:setosa
## 10 4.9:3.1:1.5:0.1:setosa
## # i 140 more rows
## # i abbreviated name:
## # 1`Sepal.Length:Sepal.Width:Petal.Length:Petal.Width:Species`
```

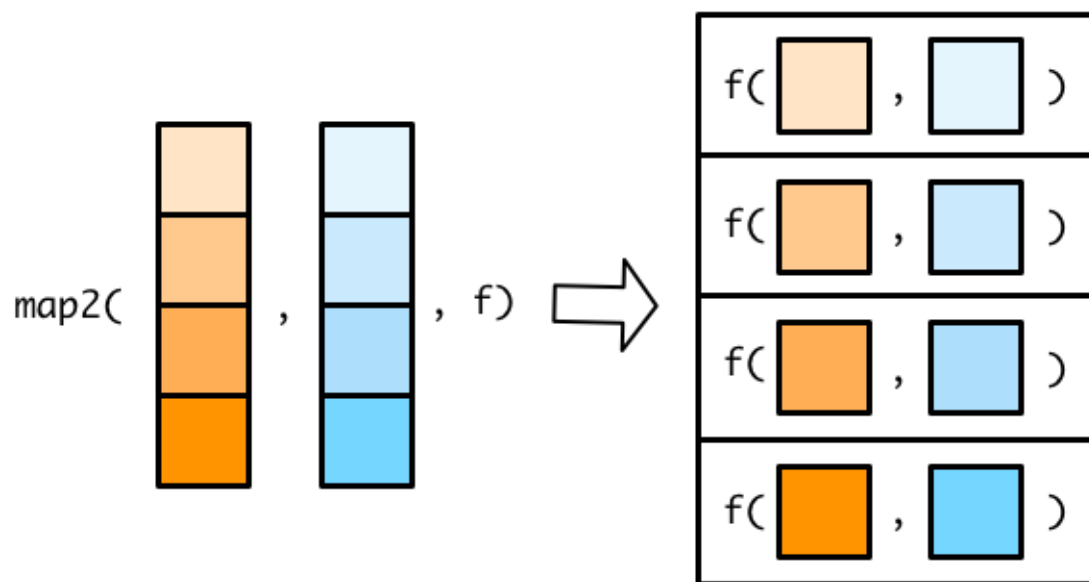
Reading in multiple files - 3rd item

```
multifile_data[[3]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1      -999          3.5          1.4          0.2 setosa
## 2      -999          3          1.4          0.2 setosa
## 3      -999          3.2          1.3          0.2 setosa
## 4       4.6          3.1          1.5          0.2 setosa
## 5       5           3.6          1.4          0.2 setosa
## 6       5.4          3.9          1.7          0.4 setosa
## 7       4.6          3.4          1.4          0.3 setosa
## 8       5           3.4          1.5          0.2 setosa
## 9       4.4          2.9          1.4          0.2 setosa
## 10      4.9          3.1          1.5          0.1 setosa
## # i 140 more rows
```

Import with different delimiters

```
delimiters <- c(",", ":", ";") # delimiters for each file  
# Write our own function to read files with specific delimiters:  
read_with_delimiter <- function(file, delimiter) {  
  read_delim(file, delim = delimiter)  
}  
  
# Map over file_list and delimiters  
multifile_data <- map2(file_list, delimiters, read_with_delimiter)
```



[\[source\]](#)

Reading in multiple files - 1st item

```
multifile_data[[1]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Reading in multiple files - 2nd item

```
multifile_data[[2]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Reading in multiple files - 3rd item

```
multifile_data[[3]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         -999         3.5         1.4         0.2 setosa
## 2         -999         3         1.4         0.2 setosa
## 3         -999         3.2         1.3         0.2 setosa
## 4          4.6         3.1         1.5         0.2 setosa
## 5          5          3.6         1.4         0.2 setosa
## 6          5.4         3.9         1.7         0.4 setosa
## 7          4.6         3.4         1.4         0.3 setosa
## 8          5          3.4         1.5         0.2 setosa
## 9          4.4         2.9         1.4         0.2 setosa
## 10         4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Summary

- `function(x){ }` or `\(x){ }` denotes a function. You also commonly see `~.x` inside `across`
- `map_df` and `modify` **apply functions** to each element of an object. `map` returns a list, `modify` returns the same object type.
- The `purrr` package has other useful functional programming features
- lists can be great for storing iterative work.

<https://sisbid.github.io/Data-Wrangling/labs/functional-program-lab.Rmd>

Extra Slides

Reading in multiple files

The `bind_rows()` function can be great for combining data.

recall that `modify` keeps the same data type (here, a list). We want a data frame instead.

See <https://www.opencasestudies.org/ocs-bp-vaping-case-study> for more information!

```
all_files_data <-  
  multiframe_data %>%  
  map_df(bind_rows, .id = "experiment")  
  
glimpse(all_files_data)
```

```
## Rows: 450  
## Columns: 6  
## $ experiment    <chr> "1", "1", "1", "1", "1", "1", "1", "1...  
## $ Sepal.Length  <dbl> 5.1, 4.9, 4.7, 4.6, 5.0, 5.4, 4.6, 5...  
## $ Sepal.Width   <dbl> 3.5, 3.0, 3.2, 3.1, 3.6, 3.9, 3.4, 3...  
## $ Petal.Length  <dbl> 1.4, 1.4, 1.3, 1.5, 1.4, 1.7, 1.4, 1...  
## $ Petal.Width   <dbl> 0.2, 0.2, 0.2, 0.2, 0.2, 0.4, 0.3, 0...  
## $ Species       <chr> "setosa", "setosa", "setosa", "setosa...
```

Shortcut for functions

Alternatively we can use `\(x)`. See this [link about function shortcuts](#).

```
my_function <- \(x){x + 1}  
my_function
```

```
## function (x)  
## {  
##     x + 1  
## }
```

```
my_function(x = my_data)
```

```
## [1] 3 4 5
```

Previously we filtered for patterns or conditions..

Dilemma: Seems a bit repetitive!

```
mtcars %>%  
  filter(cyl > 3 & cyl < 8,  
         gear > 3 & gear < 8,  
         carb > 3 & carb < 8)
```

```
##           mpg  cyl  disp  hp  drat    wt  qsec  vs  am  
## Mazda RX4      21.0    6 160.0 110  3.90  2.620 16.46  0  1  
## Mazda RX4 Wag  21.0    6 160.0 110  3.90  2.875 17.02  0  1  
## Merc 280       19.2    6 167.6 123  3.92  3.440 18.30  1  0  
## Merc 280C      17.8    6 167.6 123  3.92  3.440 18.90  1  0  
## Ferrari Dino   19.7    6 145.0 175  3.62  2.770 15.50  0  1  
##           gear carb  
## Mazda RX4           4    4  
## Mazda RX4 Wag       4    4  
## Merc 280             4    4  
## Merc 280C            4    4  
## Ferrari Dino        5    6
```

Now we can filter multiple columns!

`if_all()`: helps us filter on multiple similar conditions simultaneously!

```
mtcars %>%  
  filter(if_all(c(cyl, gear, carb), ~.x > 3 & .x < 8))
```

```
##           mpg  cyl  disp  hp  drat    wt    qsec  vs  am  
## Mazda RX4      21.0    6 160.0 110  3.90  2.620 16.46  0  1  
## Mazda RX4 Wag  21.0    6 160.0 110  3.90  2.875 17.02  0  1  
## Merc 280       19.2    6 167.6 123  3.92  3.440 18.30  1  0  
## Merc 280C      17.8    6 167.6 123  3.92  3.440 18.90  1  0  
## Ferrari Dino   19.7    6 145.0 175  3.62  2.770 15.50  0  1  
##  
##           gear  carb  
## Mazda RX4           4    4  
## Mazda RX4 Wag       4    4  
## Merc 280             4    4  
## Merc 280C            4    4  
## Ferrari Dino        5    6
```

Using across with helpers to apply function to multiple columns

?tidyr_tidy_select

```
mtcars %>%  
  mutate(across(.cols = disp:wt, round)) %>%  
  head(2)
```

```
##           mpg cyl  disp  hp  drat  wt   qsec  vs  am  gear  
## Mazda RX4      21   6  160 110    4   3  16.46   0   1    4  
## Mazda RX4 Wag  21   6  160 110    4   3  17.02   0   1    4  
##           carb  
## Mazda RX4      4  
## Mazda RX4 Wag  4
```

```
mtcars %>%  
  mutate(across(.cols = everything(), round))%>%  
  head(2)
```

```
##           mpg cyl  disp  hp  drat  wt   qsec  vs  am  gear  carb  
## Mazda RX4      21   6  160 110    4   3    16    0   1    4    4  
## Mazda RX4 Wag  21   6  160 110    4   3    17    0   1    4    4
```

Speed test! modify_if vs across

```
system.time(iris %>%  
  modify_if(is.factor, as.character))
```

```
##      user  system elapsed  
##    0.001    0.000    0.000
```

```
system.time(iris %>%  
  mutate(across(.cols = where(is.factor), as.character)))
```

```
##      user  system elapsed  
##    0.003    0.000    0.004
```

List referencing

You can also select multiple lists with the single brackets.

```
mylist[1:2] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"  
##  
## $numbers  
## [1] 1 2 3
```


Alternatively we could have fixed the second file

Looks like it had a different delimiter First, separating by the `:`.

```
multifile_data[[2]] <-  
  separate(  
    multifile_data[[2]],  
    col = 1,  
    into = colnames(multifile_data[[1]]),  
    sep = ":"  
  )
```

```
## Warning: Expected 5 pieces. Missing pieces filled with `NA` in 150  
## rows [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,  
## 16, 17, 18, 19, 20, ...].
```

```
head(multifile_data[[2]], 3)
```

```
## # A tibble: 3 × 5  
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
##   <chr>        <chr>        <chr>        <chr>        <chr>  
## 1 5.1         <NA>         <NA>         <NA>         <NA>  
## 2 4.9         <NA>         <NA>         <NA>         <NA>  
## 3 4.7         <NA>         <NA>         <NA>         <NA>
```

Fixing the second file

Second, making sure values are numeric.

```
multifile_data[[2]] <-  
  multifile_data[[2]] %>%  
  mutate(across(!Species, as.numeric))  
  
head(multifile_data[[2]], 3)
```

```
## # A tibble: 3 × 5  
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
##           <dbl>         <dbl>         <dbl>         <dbl> <chr>  
## 1           5.1           NA           NA           NA <NA>  
## 2           4.9           NA           NA           NA <NA>  
## 3           4.7           NA           NA           NA <NA>  
...  
...  
...
```